

Henry Demarest

New Haven, CT | Tarrytown, NY | 914.621.1424 | henry.demarest@yale.edu | www.henrydemarest.me

Education

Yale University: New Haven, Connecticut (*August 2021 - May 2025*); GPA: 3.97

- **Major(s):** Mechanical Engineering (ABET), BS Computer Science
- **Relevant Mechanical Engineering Coursework:** Thermodynamics, Fluid Mechanics, Material Science, Statics, Mechatronics, Mechanical Design, Machine Design, Computer-Aided Design, Linear Algebra, Differential Equations
- **Relevant Computer Science Coursework:** Data Structures, Computer System Architecture, Algorithms, Intelligent Robotics
- **Non-Technical Groups/Activities:** Mechanical Engineering Peer Mentor, Engineering Tour Guide, Yale Club Running, DPods Orchestra Violist, Freshman Orientation Backpacking Trip Leader, Intramural Sports Secretary

Irvington High School: Irvington, New York (*September 2017 - June 2021*); Weighted GPA = 4.58; SAT = 1600

- **Awards:** Valedictorian (2021), National Merit Semifinalist (2021), U.S. Presidential Scholar Semi-Finalist (2021)
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Experience

Astrobotic Technologies – Mechanical Engineering Intern (*Summer 2023*):

- Designed fixture for vacuum testing of a Roll-Out Solar Array (ROSA) in cryogenic & elevated temperatures using 304 SS
- Iterated upon the design of a spacerless MLI thermal enclosure to enable spacecraft survival of the lunar night
- Developed, manufactured, and tested a motorized test fixture for cryogenic fabric testing for in-space robotic applications
- Created CAD models in consultation with a customer for the production of renderings of a custom lunar rover design
- Communicated with part vendors/customers, presented to project managers, and created documentation for my designs

Yale Undergraduate Aerospace Association – Executive Board President (*2023 - Present*):

Project Liquid – Chief Engineer (*2023 - Present*), **Co-President** (*2022 - 2023*), **Propulsion Lead** (*2021 - 2022*):

- Lead technical development of a bipropellant liquid rocket engine designed for use with Nitrous Oxide and Ethane
- Coordinate and assist subteams to ensure design/manufacturing progress meets deadlines and technical objectives
- Teach other club members about concepts related to fluid mechanics, 3D design, and rocket engine operation
- Communicate with teams at other schools, research previous engine designs, and learn from industry contacts
- Led design of a coaxial shear injector that used a stacked manifold design as the Propulsion Lead

CubeSat Team – Attitude Determination and Control System (ADCS) Lead (*2022 - 2023*), **Member** (*2021-2022*):

- Led the ADCS Subteam to prepare CubeSat for launch as part of NASA's CubeSat Launch Initiative Award
- Designed algorithm to control CubeSat's active stabilization efforts (see research below)
- Collaborated on the design of an attachment system for the Gravity Gradient Boom (GGB)

Yale University – Undergraduate Learning Assistant for Computer Science Course (*2022 - 2023*):

- Worked as a member of the teaching staff for CPSC 223 (Data Structures & Programming Techniques)
- Held office hours and taught students topics ranging from specific programming syntax to general problem solving strategies

Research

Yale Nanoprobe Research Group – Researcher (*2023 - Present*):

- Conducting research on the microstructural effects on bulk properties of aging MP35N, a nickel-chromium superalloy
- Using tools, such as nanoindentation, scanning electron microscopy, x-ray diffraction, and other material analysis techniques

Cube Satellite Detumbling Research Project – Researcher (*Summer 2022*):

- Spent 10 weeks working on a detumbling algorithm for a 2U CubeSat with a limited suite of sensors and control hardware
- Used TRIAD algorithm for attitude determination and B-dot controllers paired with magnetorquers for attitude correction
- Created MATLAB simulations to experiment with various control system parameters and assess algorithm performance

Machine Learning Science Research Project – Student Researcher with Mentors at IBM (*2019 - 2021*):

- Assessed the performance of different reinforcement learning algorithms with a bipedal walking task using Python code
- Earned 1st Place at the NYS Science and Engineering Fair (NYSSEF) and attended the Regeneron ISEF competition

Volunteering

Fuller Center for Housing – High School Chapter Founder, Volunteer Supervisor, Intern (*2018 - 2021*):

- Worked on a construction crew to repair and remodeled local homes to be used as affordable housing in local communities
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Skills

Computer Programming: C, C++, Java, Python, MATLAB

CAD: Solidworks, Autodesk Inventor/Fusion 360, OnShape

Manufacturing: Lathe, Mill, CNC, 3D-Printer, Hand Tools

Software: Project Organization (ClickUp, Slack, Confluence),

Project Sharing (GrabCAD, GitHub), Microsoft Office

Language: Conversational Spanish